# Supplemental Material: Proofs of Theoretical Properties for "Zero-Trust Shielded Graph-Structured Decision Control for Secure Autonomous Recovery in AI-Native 6G Disaster-Monitoring Sensing Networks"

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This supplemental document provides the complete proofs for the ten propositions stated in the main paper. All proposition numbers and references correspond to those in the main text.

*Proof of Proposition 1 (Monotone conservatism).* Since $w_1, w_2, w_3, w_4 > 0$, the composite gate score $g_t = w_1(1-\tau_t) + w_2\rho_t + w_3 u_t + w_4\Delta_t$ is monotonically decreasing in $\tau_t$ and increasing in $\rho_t, u_t, \Delta_t$. Since $\kappa_1, \kappa_2 \geq 0$, $\tau_{\min} = \tau_0 + \kappa_1\rho_t + \kappa_2 u_t$ is non-decreasing in $\rho_t, u_t$, making the trust condition harder to satisfy. The ALLOW branch requires $g_t < \gamma_1$, $\tau_t \geq \tau_{\min}$, and $\Delta_t \leq \delta_{\max}$. Worsening any input can only increase $g_t$, raise $\tau_{\min}$, or violate the divergence bound, pushing the output toward a more conservative branch. □

*Proof of Proposition 2 (Unauthorized-actuation exclusion).* In the shield algorithm, $\alpha_t = 0$ triggers $d_t = \text{BLOCK}$ at the authorization check, before any permissive branch is evaluated. □

*Proof of Proposition 3 (Divergence-triggered safeguard).* $\Delta_t > \delta_{\max}^{\text{hard}}$ triggers $d_t = \text{BLOCK}$ at the hard-divergence check. $\Delta_t > \delta_{\max}$ violates the ALLOW condition, preventing permissive decisions. □

*Proof of Proposition 4 (Bounded decision completeness).* The shield algorithm is a finite ordered branching structure with mutually exclusive branches (strict threshold ordering) and a final ELSE clause catching all remaining cases. Every valid input therefore maps to exactly one $d_t \in \mathcal{D}$. □

*Proof of Proposition 5 (Per-decision complexity).* Graph encoding: $\mathcal{O}(L|E_t|d)$. Candidate scoring and trust–risk evaluation: $\mathcal{O}(|\mathcal{A}|k)$ each. Shield evaluation: $\mathcal{O}(1)$. Total: $\mathcal{O}(L|E_t|d + |\mathcal{A}|k)$. □

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*Proof of Proposition 6 (Defense-in-depth bound).* Under identity attack, an unsafe ALLOW requires an authorization false negative: $\Pr \leq \varepsilon_\alpha$. Under telemetry attack: $\Pr \leq \varepsilon_\tau$. Under combined attack with conditional independence: $\Pr[\text{ALLOW} \mid \text{combined}] = \Pr[\alpha_t = 1 \mid \text{id}] \cdot \Pr[g_t < \gamma_1 \mid \text{telem}] \leq \varepsilon_\alpha \varepsilon_\tau$. □

*Proof of Proposition 7 (Calibration-consistent accuracy).* Each episode produces Bernoulli $y_i$ with $\mathbb{E}[y_i] = \varepsilon^\star$. By Hoeffding's inequality: $\Pr[\varepsilon^\star - \hat{\varepsilon} > t] \leq \exp(-2nt^2)$. Setting $\exp(-2nt^2) = \delta$ gives $t = \sqrt{\ln(1/\delta)/(2n)}$. □

*Proof of Proposition 8 (Bounded price of safety).* With probability $(1 - \eta - \eta_{\text{block}})$, the shield allows (loss 0). With probability $\eta$, scope-reduction yields loss $\leq U_{\text{gap}}$. With probability $\eta_{\text{block}}$, blocking yields loss $\leq U_{\max}$. Summing: $\mathbb{E}[\text{loss}] \leq \eta U_{\text{gap}} + \eta_{\text{block}} U_{\max}$. □

*Proof of Proposition 9 (Conformal admissibility coverage).* By the conformal framework [?], if scores $(r_1, \ldots, r_{n+1})$ are exchangeable, $\Pr[r_{n+1} \leq r_{(\lceil(1-\alpha)(n+1)\rceil)}] \geq 1 - \alpha$. The shield allows only if $g_t < \gamma_1$, which corresponds to $r_{n+1} > r_{(\lceil(1-\alpha)(n+1)\rceil)}$, occurring with probability $\leq \alpha$. □

*Proof of Proposition 10 (Calibration convergence rate).* By Hoeffding's concentration, $|\hat{\varepsilon}(\gamma) - \varepsilon(\gamma)| \leq \sqrt{\ln(2/\delta)/(2n)}$ w.h.p. Since $\varepsilon'(\gamma) \geq c_\varepsilon > 0$, the mean value theorem gives $|\varepsilon(\gamma_1^{(n)}) - \varepsilon(\gamma_1^\star)| \geq c_\varepsilon |\gamma_1^{(n)} - \gamma_1^\star|$. The triangle inequality yields $c_\varepsilon |\gamma_1^{(n)} - \gamma_1^\star| = \mathcal{O}(1/\sqrt{n})$, so $|\gamma_1^{(n)} - \gamma_1^\star| = \mathcal{O}_P(1/\sqrt{n})$. □

## References

[1] A. N. Angelopoulos and S. Bates, "A gentle introduction to conformal prediction and distribution-free uncertainty quantification," *Foundations and Trends in Machine Learning*, vol. 16, no. 4, pp. 494–591, 2023.